

EOD Report – Git Challenges

Date: 21 August 2026

Topic: Advanced Git Challenges

Status: Completed / Practiced

Objective

The objective of today's task was to practice advanced Git and GitHub operations, understand how to recover from common Git mistakes, and learn different collaboration workflows.

Tasks Completed

1. Resolve Merge Conflicts

- Created a merge conflict intentionally by modifying the same line in two branches.
- Merged the branches and identified the conflict.
- Resolved the conflict manually.
- Added the resolved file and committed the changes.
- Pushed the final changes to GitHub.

Commands practiced:

```
git checkout -b feature
```

```
git checkout -b master
```

```
git merge feature-branch
```

```
git status
```

```
git add .
```

```
git commit
```

```
git push
```

```

C:\Users\Zeehan>git merge feature-branch
Auto-merging file.txt
CONFLICT (add/add): Merge conflict in file.txt
CONFLICT (modify/delete): file.txt deleted in feature-branch and modified in HEAD. Version HEAD of file.txt is newer.
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\Zeehan>git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

changes to be committed:
  deleted:    file2.txt
  deleted:    file3.txt
  deleted:    file4.txt
  deleted:    sample.txt

Unmerged paths:
  (use "git add/rm <file>..." as appropriate to mark resolution)
  both added: file.txt
  deleted by them: file.txt

C:\Users\Zeehan>git merge
5

```

```
hp@Zeeshan:~/coding/hiring-app$  
deleted:    file2.txt  
deleted:    file3.txt  
deleted:    file4.txt  
deleted:    sample.txt  
  
hp@Zeeshan MINGW64 ~/hiring-app (master|MERGING)  
$ git commit -m "during merge"  
[master c98dd08] during merge  
  
hp@Zeeshan MINGW64 ~/hiring-app (master)  
$ ll  
total 5  
-rw-r--r-- 1 hp 197121 21 Aug 21 12:07 README.md  
-rw-r--r-- 1 hp 197121 9 Aug 21 12:07 another.txt  
-rw-r--r-- 1 hp 197121 32 Aug 21 12:07 credential.txt  
-rw-r--r-- 1 hp 197121 133 Aug 21 12:09 file.txt  
-rw-r--r-- 1 hp 197121 35 Aug 21 12:07 file1.txt  
-rw-r--r-- 1 hp 197121 0 Aug 21 12:07 webhook-test.txt  
  
hp@Zeeshan MINGW64 ~/hiring-app (master)  
$ cat file.txt  
<<<<<< HEAD  
Welcome to my page from "master-branch"  
  
Welcome to my project from feature-branch  
feature-branch  
  
hp@Zeeshan MINGW64 ~/hiring-app (master)  
$
```

2. Recover Deleted Branch

- Created a local branch and committed changes.
- Deleted the branch.
- Used git reflog to find the previous commit.
- Recreated the deleted branch using the commit ID.

Commands:

```
git branch -d feature2
```

git relog

git checkout -b feature2 <commit-id>

3. Undo Wrong Push

- Created and pushed an incorrect commit.
- Used git revert

Commands:

git revert <commit-id>

4. Amend a Commit

- Created a commit without a required file.
- Added the missing file.
- Used git commit --amend to update the previous commit.

Commands:

git add missing-file

git commit --amend

```
hp@Zeeshan MINGW64 ~/hirring-app
new file mode 100644
index 0000000..d36c7ec
--- /dev/null
+++ b/file02
@@ -0,0 +1 @@
+welcome co my file
diff --git a/file03 b/file03
new file mode 100644
index 0000000..2e10881
--- /dev/null
+++ b/file03
@@ -0,0 +1 @@
+welcome 02

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git commit --amend -m "Add file2 and file3"
[feature-branch 78e66a8] Add file2 and file3
Date: Fri Aug 21 13:05:43 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 file02
create mode 100644 file03

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git log --oneline
78e66a8 (HEAD -> feature-branch) Add file2 and file3
d9ac91a Revert "testing revert"
f9c5b97 (origin/feature-branch) testing revert
87cae74 add file master

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ |

hp@Zeeshan MINGW64 ~/hirring-app
Commit: 33085de334035cf2268594b0118eb39cfff605 (HEAD -> feature-branch)
Author: zeeshan123 <zeeshan123>
Date: Fri Aug 21 13:05:43 2026 +0530

Add file1 and file2

diff --git a/file02 b/file02
new file mode 100644
index 0000000..d36c7ec
--- /dev/null
+++ b/file02
@@ -0,0 +1 @@
+welcome co my file
diff --git a/file03 b/file03
new file mode 100644
index 0000000..2e10881
--- /dev/null
+++ b/file03
@@ -0,0 +1 @@
+welcome 02

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git commit --amend -m "Add file2 and file3"
[feature-branch 78e66a8] Add file2 and file3
Date: Fri Aug 21 13:05:43 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 file02
create mode 100644 file03

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ |
```

5. Cherry-pick a Commit

- Created a commit on one branch.
- Switched to another branch.
- Applied only the required commit using git cherry-pick.

Command:

git cherry-pick <commit-id>

```
hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git switch master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

$ git log --oneline
70e4f41 (HEAD -> master, origin/master, origin/HEAD) Merge branch 'master' of https://github.com/Zeeshancloud15/hirring-app
dbefb69 add file in feature
87cae74 add file master
e6d070c (origin/main) Merge branch 'master' of https://github.com/Zeeshancloud15/hirring-app
81f9eb4 webhook
8c7e1a0 add two sample file
e80e3e4 pulling
238aa0b Merge pull request #1 from Zeeshancloud15/hotfix
f443176 update file1.txt update file1.txt in hotfix
603355e hide credential
f8e4a8d Create .gitignore
0fec8e2 Add initial content to README.md
41f200c add in sample file in feature-branch
4a7f6ee add two sample file
ca9d738 new files added

hp@Zeeshan MINGW64 ~/hirring-app (master)
$ |
```

```
new file mode 100644
index 0000000..d36c7ec
--- /dev/null
+++ b/file02
@@ -0,0 +1 @@
+welcome co my file
diff --git a/file03 b/file03
new file mode 100644
index 0000000..2e10881
--- /dev/null
+++ b/file03
@@ -0,0 +1 @@
+welcome 02

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git commit --amend -m "Add file2 and file3"
[feature-branch 78e66a8] Add file2 and file3
Date: Fri Aug 21 13:05:43 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 file02
create mode 100644 file03

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ git log --oneline
78e66a8 (HEAD -> feature-branch) Add file2 and file3
d9ac91a Revert "testing revert"
f9c5b97 (origin/feature-branch) testing revert
87cae74 add file master

hp@Zeeshan MINGW64 ~/hirring-app (feature-branch)
$ |
```

6. Interactive Rebase

- Created multiple commits.
- Used interactive rebase to reorder and squash commits.
- Combined multiple commits into one clean commit.

Command:

git rebase -i HEAD~3

```

[detached HEAD 922cae1] add file in feature
Date: Fri Aug 21 13:22:39 2026 +0530
4 files changed, 7 insertions(+)
create mode 100644 file02
create mode 100644 file03
Successfully rebased and updated refs/heads/master.

hp@Zeeshan MINGW64 ~/hiring-app (master)
$ git log --oneline
922cae1 (HEAD -> master) add file in feature
35cd410 # This is a combination of 8 commits. # This is the 1st commit message:
67cae74 add file master

hp@Zeeshan MINGW64 ~/hiring-app (master)
$
```

```

pick dbeffb69 # add file in feature
squash ea9d738 # new files added
squash 6c7e1a0 # add two sample file
squash 8719eb4 # webhook
squash 4a7f6ee # add two sample file
squash 41f200c # add in sample file in feature-branch
squash 603355e # hide credential
squash 0fec8e2 # Add initial content to README.md
squash fb4e48d # Create .gitignore
squash 3f4917b # Update file1.txt update file1.txt in hotfix
squash f0efba6 # Add file2 and file3

# Rebase 67cae74..f0efba6 onto 87cae74 (11 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-c | -C] <commit> = "like "squash" but keep only the previous
#   commit's log message, unless -C is used, in which case
#   keep only this commit's message; -C is same as -c but
#   opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C <commit> | -c <commit>] <label> {<tag> | <branch>}
git/rebase=merge/git-rebase-todo[*] (unix) (13:19 21/08/2026) 11,7 Top
-- INSERT --
```

7. Tagging & Release

- Created a version tag v1.0.
- Pushed the tag to GitHub.
- Practiced deleting and recreating the tag.

Commands:

git tag v1.0

git push origin v1.0

git tag -d v1.0

git push origin --delete v1.0

git tag v1.0 <commit-id>

git push origin v1.0

```

hp@Zeeshan MINGW64 ~/hiring-app (master)
$ git log --oneline
922cae1 (HEAD -> master, tag: v1.0) add file in feature
35cd410 # This is a combination of 8 commits. # This is the 1st commit message:
67cae74 add file master

hp@Zeeshan MINGW64 ~/hiring-app (master)
$
```

8. Clone with Sparse Checkout

- Practiced cloning a repository while checking out only the required subdirectory.
- Used Git sparse checkout.

Commands:

```
git clone --no-checkout <repository-url>
cd <repository>
git sparse-checkout init --cone
git sparse-checkout set <directory>
git checkout
```

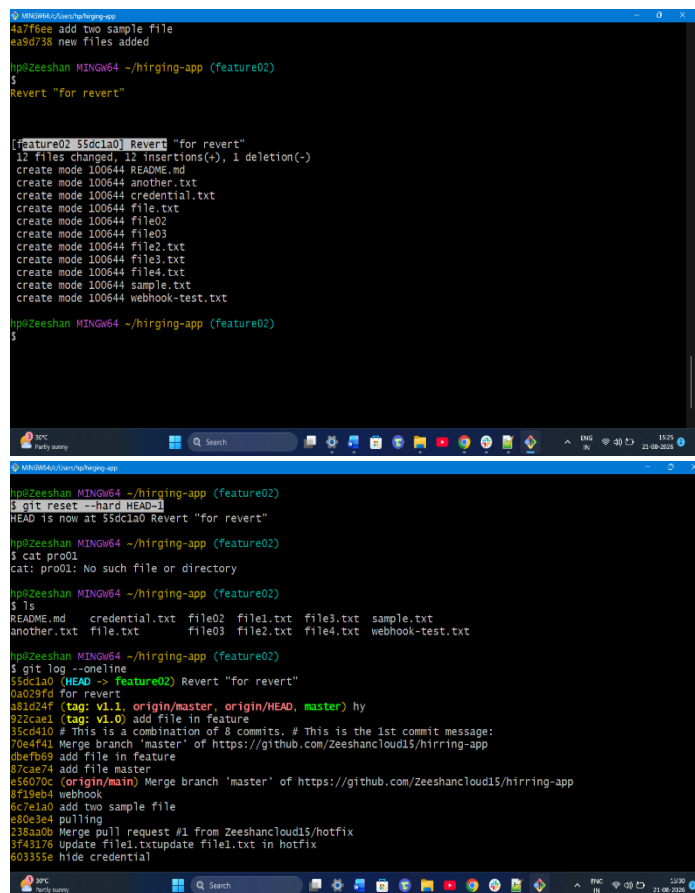
9. Reset vs Revert

- Practiced both git reset --hard and git revert.
- Understood that reset moves the branch pointer and can remove local history from the current branch.
- Understood that revert creates a new commit that reverses an earlier commit.

Commands:

```
git reset --hard HEAD~1
```

```
git revert <commit-id>
```



```
MINGW64-Shell-PowerShell
4a7f6ee add two sample file
ea9d738 new files added

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$
Revert "for revert"

(feature02 55dc1a0) Revert "for revert"
12 files changed, 12 insertions(+), 1 deletion(-)
create mode 100644 README.md
create mode 100644 another.txt
create mode 100644 credential.txt
create mode 100644 file1.txt
create mode 100644 file02
create mode 100644 file03
create mode 100644 file2.txt
create mode 100644 file3.txt
create mode 100644 file4.txt
create mode 100644 sample.txt
create mode 100644 webhook-test.txt

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$ git reset --hard HEAD~1
HEAD is now at 55dc1a0 Revert "for revert"

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$ cat pro01
cat: pro01: No such file or directory

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$ ls
README.md  credential.txt  file02  file1.txt  file3.txt  sample.txt
another.txt  file.txt      file03  file2.txt  file4.txt  webhook-test.txt

hp8Zeeshan MINGW64 ~/hiring-app (feature02)
$ git log --oneline
55dc1a0 (HEAD -> feature02) Revert "for revert"
0a029fd for revert
a81d24f (tag: v1.1 origin/master origin/HEAD, master) hy
922c8e1 (tag: v1.0) add file in feature
35cd410 # This is a combination of 8 commits. # This is the 1st commit message:
70e4f41 Merge branch 'master' of https://github.com/Zeeshancloud15/hiring-app
dbef0b9 add file in feature
07cae74 add file master
a56070c (origin/main) Merge branch 'master' of https://github.com/Zeeshancloud15/hiring-app
8f19eb4 webhook
6c7e1a0 add two sample file
e80e3e4 pulling
138aa0b Merge pull request #1 from Zeeshancloud15/hotfix
3f43176 Update file1.txtupdate file1.txt in hotfix
603359e hide credential
```

10. Detached HEAD

- Checked out a specific commit instead of a branch.
- Observed the detached HEAD state.
- Created a new branch from that commit to preserve the work.

Commands:

```
git checkout <commit-id>
```

```
git status
```

```
git checkout -b new-branch
```

```
Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at d9ac91a Revert "testing revert"

hp@Zeeshan MINGW64 ~/hirging-app ((d9ac91a...))
$ git branch detach-branch

hp@Zeeshan MINGW64 ~/hirging-app ((d9ac91a...))
$ git switch detach-branch
Switched to branch 'detach-branch'

hp@Zeeshan MINGW64 ~/hirging-app (detach-branch)
$ git log --oneline -3
d9ac91a (HEAD -> detach-branch) Revert "testing revert"
79c5b97 (origin/feature-branch) testing revert
07cae74 add file master

hp@Zeeshan MINGW64 ~/hirging-app (detach-branch)
$ git branch
* detach-branch
  feature-branch
  feature02
  master
  recover

hp@Zeeshan MINGW64 ~/hirging-app (detach-branch)
$
```

11. Git Hooks

- Created a pre-commit hook.
- Configured the hook to validate the commit message format.
- Practiced rejecting commits that do not start with a Jira format such as JIRA-123.

Example format:

JIRA-123 Fix login issue

Hook location:

.git/hooks/pre-commit

```
hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ git add .
warning: in the working copy of 'test0.txt', LF will be replaced by CRLF the next time Git touches it

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ git commit -m "jira-xxx Add test file"
ERROR: Commit message must start with JIRA-XXX

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ git commit -m "jira-789 Add test file"
ERROR: Commit message must start with JIRA-XXX

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ git commit -m "jira-789 Add test file"
ERROR: Commit message must start with JIRA-XXX

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ git commit -m "JIRA-789 Add test file"
[feature02 7388de5] JIRA-789 Add test file
1 file changed, 1 insertion(+)
create mode 100644 test0.txt

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ |

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ ls
README.md  credential.txt  file02  file1.txt  file3.txt  sample.txt  test0.txt
another.txt  file.txt  file03  file2.txt  file4.txt  test.txt  webhook-test.txt

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ ls -a
./  ../  .git/  README.md  file.txt  file1.txt  file4.txt  test0.txt
.file.txt.swp  .gitignore.swp  another.txt  file02  file2.txt  sample.txt  webhook-test.txt
credential.txt  file03  file3.txt  test.txt

hp@Zeeshan MINGW64 ~/hirging-app (feature02)
$ cd .git

hp@Zeeshan MINGW64 ~/hirging-app/.git (GIT_DIR!)
$ ls
COMMIT_EDITMSG  HEAD  REBASE_HEAD  description  index  logs/  refs/
FETCH_HEAD  ORIG_HEAD  config  hooks/  info/  objects/

hp@Zeeshan MINGW64 ~/hirging-app/.git (GIT_DIR!)
$ cd hooks

hp@Zeeshan MINGW64 ~/hirging-app/.git/hooks (GIT_DIR!)
$ ls
applypatch-msg.sample*  post-update.sample*  pre-push.sample*  push-to-checkout.sample*
commit-msg.sample*  pre-applypatch.sample*  pre-rebase.sample*  sendemail-validate.sample*
commit-msg.sample*  pre-commit.sample*  pre-receive.sample*  update.sample*
fsmonitor-watchman.sample*  pre-merge-commit.sample*  prepare-commit-msg.sample*

hp@Zeeshan MINGW64 ~/hirging-app/.git/hooks (GIT_DIR!)
$
```

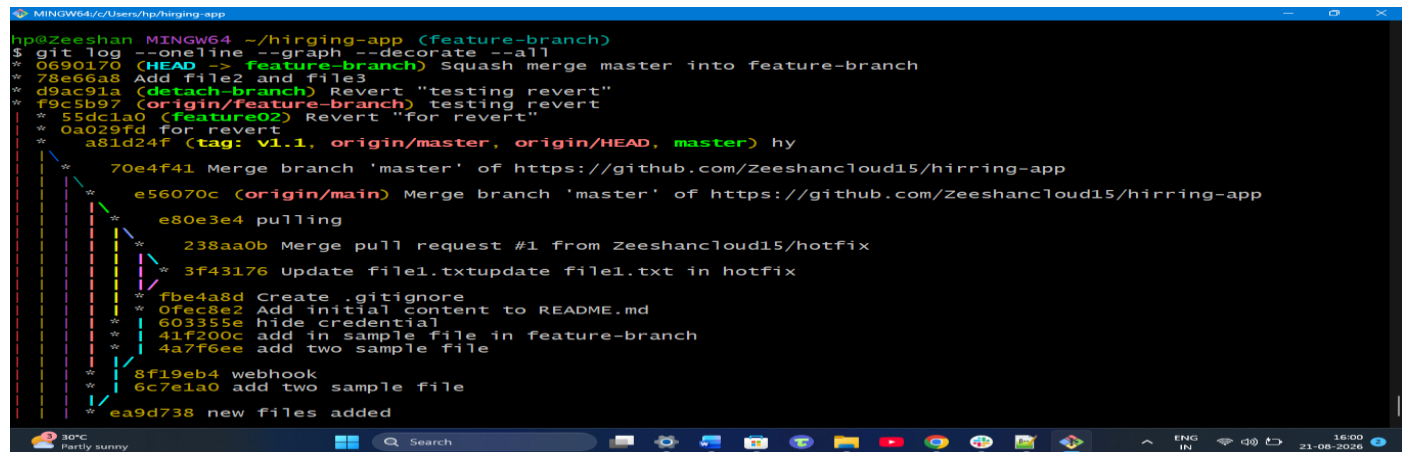
12. Squash Merge vs Rebase Merge

- Practiced squash merging to combine multiple feature commits into one commit.
- Practiced rebasing to apply commits on top of the latest target branch.
- Compared the resulting Git history.

Commands practiced:

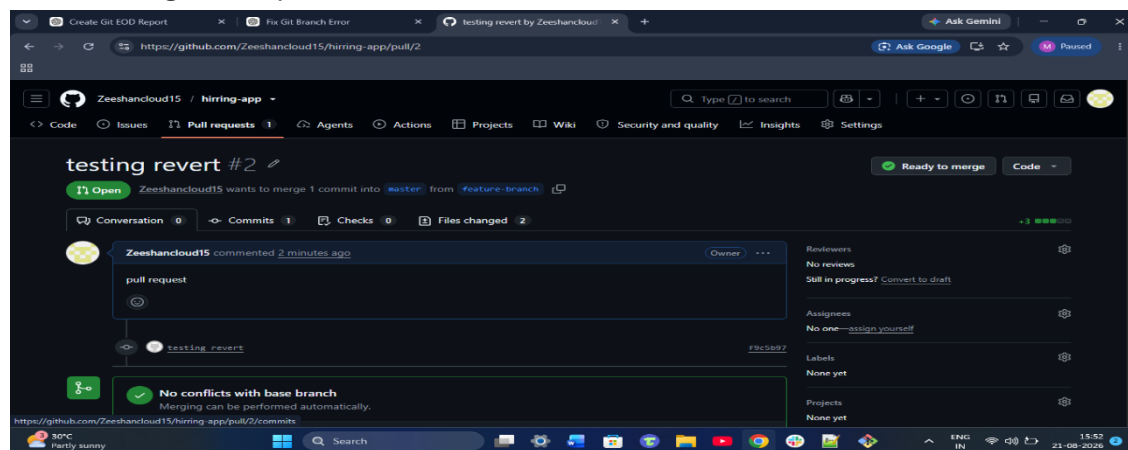
git merge --squash feature

git rebase main



13. Fork & Pull Request Workflow

- Practiced the GitHub fork workflow.
- Created a fork of a repository.
- Cloned the fork locally.
- Created a feature branch.
- Made changes and pushed the branch.



14. Recover Lost Commit

- Created a commit.
- Used git reset --hard to move the branch backward.
- Used git reflog to locate the lost commit.
- Recovered the commit using its commit ID.

Commands:

git reset --hard HEAD~1

git reflog

git checkout -b recovered <commit-id>

```
MINGW64/C:/Users/hp/hirging-app
a81d24f (tag: v1.1, origin/master, origin/HEAD, master) HEAD@{9}: commit (merge): hy
922cae1 (tag: v1.0) HEAD@{10}: rebase (finish): returning to refs/heads/master
922cae1 (tag: v1.0) HEAD@{11}: rebase (squash): add file in feature
66e9f83 HEAD@{12}: rebase (squash): # This is a combination of 10 commits.
12b11f7 HEAD@{13}: commit: rebase
35cd410 HEAD@{14}: rebase (squash): # This is a combination of 8 commits.
11f0884 HEAD@{15}: rebase (squash): # This is a combination of 7 commits.
c214e7e HEAD@{16}: rebase (squash): # This is a combination of 6 commits.
d8f4d8e HEAD@{17}: rebase (squash): # This is a combination of 5 commits.
ec0924d HEAD@{18}: rebase (squash): # This is a combination of 4 commits.
2c3a0d3 HEAD@{19}: rebase (squash): # This is a combination of 3 commits.
54fa206 HEAD@{20}: rebase (squash): # This is a combination of 2 commits.
dbefb69 HEAD@{21}: rebase (start): checkout HEAD~3
f0efba6 HEAD@{22}: cherry-pick: Add file2 and file3
70e4f41 HEAD@{23}: checkout: moving from feature-branch to master
78e66a8 (feature-branch) HEAD@{24}: commit (amend): Add file2 and file3
330b5db HEAD@{25}: commit (amend): Add file1 and file2
defd23e HEAD@{26}: commit: add file02
d9ac91a (HEAD -> detach-branch) HEAD@{27}: revert: Revert "testing revert"
f9c5b97 (origin/feature-branch) HEAD@{28}: commit: testing revert

hp@Zeeshan MINGW64 ~/hirging-app (detach-branch)
$ git branch
* detach-branch
  feature-branch
  feature02
  master
  recover

hp@Zeeshan MINGW64 ~/hirging-app (detach-branch)
$
```

Conclusion

Today's Git challenge helped me understand advanced Git commands and real-world Git workflows. I learned how to resolve conflicts, recover deleted or lost work, modify commits, transfer specific commits, maintain clean history, create releases, and collaborate through GitHub Pull Requests.